Book 1 — Lecture 2 — Exercise 6

Period of Simple Harmonic Motion

Prompt. How long does it take for the oscillating particle to go through one full cycle of motion?[1]

Sinusoidal form

Let $x(t)$ be the displacement from equilibrium. In SHM one may write

$$x(t) = A\cos(\omega t + \phi) \qquad \text{or} \qquad x(t) = A\sin(\omega t + \phi'),$$

with constant amplitude $A > 0$, angular frequency $\omega > 0$, and phase ϕ (or ϕ').[2]

One full cycle

A full cycle means the particle returns to the same *state*: same position *and* same direction of motion (same value of dx/dt). For the cosine form, that first recurrence after time $T > 0$ occurs when the argument increases by 2π:

$$\omega(t + T) + \phi = \omega t + \phi + 2\pi \quad \Longrightarrow \quad \omega T = 2\pi.$$

Hence the *period* (time for one complete cycle) is

$$T = \frac{2\pi}{\omega}.$$

Spring–mass form

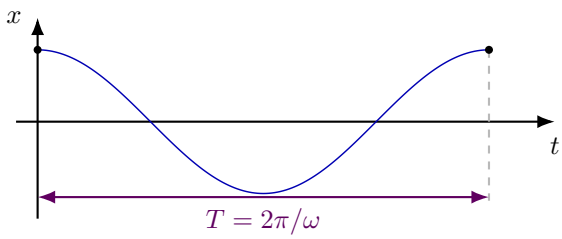
For a mass m on a spring of constant k (ideal Hooke's law near equilibrium), $\omega = \sqrt{k/m}$, so

$$T = 2\pi\sqrt{\frac{m}{k}}.$$

Picture. One period is one "hump" of $x(t)$ from crest to crest (or trough to trough), or—equivalently—any time interval over which the phase $\omega t + \phi$ advances by 2π.

[1] "Simple harmonic motion" means motion governed by a restoring force linear in displacement from equilibrium, producing sinusoidal position vs. time when the coordinate is measured from the equilibrium point.

[2] The two forms differ only by a phase shift of $\pi/2$ between sin and cos; either yields the same period.



Result. The time for one full cycle of SHM is the period

$$T = \frac{2\pi}{\omega}.$$

For a spring–mass oscillator, $\omega = \sqrt{k/m}$, so $T = 2\pi\sqrt{m/k}$.

Example. A stiff spring (large k) shortens the period; a heavier mass (longer m) lengthens it.